spectrum. In comparison, taking the Fourier transform of an image converts the straightforward information in the spatial domain into a scrambled form in the frequency domain. In short, don't expect the Fourier transform to help you understand the information encoded in images.

Likewise, don't look to the frequency domain for filter design. The basic feature in images is the edge, the line separating one *object* or *region* from another *object* or *region*. Since an edge is composed of a wide range of frequency components, trying to modify an image by manipulating the frequency spectrum is generally not productive. Image filters are normally designed in the spatial domain, where the information is encoded in its simplest form. Think in terms of *smoothing* and *edge enhancement* operations (the spatial domain) rather than *high-pass* and *low-pass* filters (the frequency domain).

In spite of this, Fourier image analysis does have several useful properties. For instance, *convolution* in the spatial domain corresponds to *multiplication* in the frequency domain. This is important because multiplication is a simpler mathematical operation than convolution. As with one-dimensional signals, this property enables FFT convolution and various deconvolution techniques. Another useful property of the frequency domain is the *Fourier Slice Theorem*, the relationship between an image and its projections (the image viewed from its sides). This is the basis of *computed tomography*, an x-ray imaging technique widely used medicine and industry.

The frequency spectrum of an image can be calculated in several ways, but the FFT method presented here is the only one that is practical. The original image must be composed of N rows by N columns, where N is a power of two, i.e., 256, 512, 1024, etc. If the size of the original image is not a power of two, pixels with a value of zero are added to make it the correct size. We will call the two-dimensional array that holds the image the **real array**. In addition, another array of the same size is needed, which we will call the **imaginary array**.

The recipe for calculating the Fourier transform of an image is quite simple: take the one-dimensional FFT of each of the rows, followed by the one-dimensional FFT of each of the columns. Specifically, start by taking the FFT of the N pixel values in row 0 of the real array. The real part of the FFT's output is placed back into row 0 of the real array, while the imaginary part of the FFT's output is placed into row 0 of the imaginary array. After repeating this procedure on rows 1 through $N-1$, both the real and imaginary arrays contain an intermediate image. Next, the procedure is repeated on each of the *columns* of the intermediate data. Take the N pixel values from column 0 of the real array, *and* the N pixel values from column 0 of the imaginary array, and calculate the FFT. The real part of the FFT's output is placed back into column 0 of the real array, while the imaginary part of the FFT's output is placed back into column 0 of the imaginary array. After this is repeated on columns 1 through $N-1$, both arrays have been overwritten with the image's frequency spectrum.